

COMP364 Quiz 6

Question 1. Transformer (5 points)

List the major quantities and components of a transformer neural network architecture, in their order of computation.

Solution.

token embedding, position embedding, input representation, a sequence of attention layers, logits via unembedding, token probabilities via softmax, token prediction.

See appendix A for some optional additional explanation of this computation.

Question 2. Transformer hyper-parameter (5 points)

Name and describe the most fundamental hyperparameter of a transformer. Give an example of a typical value of this hyperparameter from a recent open-weights model.

Solution.

The *model dimension* d_{model} is the length of vectors that represent input and output in the transformer. Intermediate computations or “states of mind” are also represented by vectors of this length. Example: The 2025 model Qwen3-32B has $d_{\text{model}} = 5120$. (An approximate number is accepted as correct.)

Question 3. Attention layer (5 points)

List the six main components in an attention layer, in the order of computation.

Solution.

layer normalization, masked self-attention, residual add, layer normalization, feed-forward network, residual add.

Question 4. Attention layer hyper-parameters (10 points)

List the two most important hyperparameters of an attention layer. Give a brief explanation of each, and a typical example from a recent open-weights model.

Solution.

- In the masked self-attention component, the *number of attention heads* H determines the *head dimension* $d_H = d_{\text{model}}/H$ (so H must be a divisor of d_{model}). We think of H and d_H as a single hyperparameter because the choice of one determines the other. Example: The 2025 model Qwen3-32B has $H = 64$ and thus $d_H = 5120/64 = 80$. (Approximate numbers are accepted as correct.)
- In the feed-forward network, the *FFN hidden dimension* d_{ff} is the number of units in the single hidden layer of the FFN. Example: The 2025 model Qwen3-32B has $d_{\text{ff}} = 5d_{\text{model}} = 25,600$. (An approximate number is accepted as correct.)

Question 5. Masked self-attention (5 points)

In the phrase *masked self-attention*, briefly describe the meaning of “masked” and “self”.

Solution.

“Masked” means that tokens cannot attend to future tokens—the future is *masked*. “Self” means that a token may attend to itself.

Question 6. Multi-head attention operator (5 points)

Describe the difference between single-head and a multi-head attention in one sentence.

Solution.

Multi-head attention splits the input representation by projecting it into H smaller representations, computing standard attention on each, then concatenating the results to produce the projected attention output.

Question 7. One-hot encoding (5 points)

Suppose we are using a vocabulary of 5 tokens, and we wish to encode the following sequence of three token IDs: $\text{seq} = (4, 2, 4)^\top$. What is the resulting one-hot token matrix?

Solution.

Token ID 4 encodes as $(0, 0, 0, 0, 1)$. Token ID 2 encodes as $(0, 0, 1, 0, 0)$. So the sequence of IDs $(4, 2, 4)^\top$ encodes as

$$\begin{pmatrix} 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix}.$$

Question 8. Embedding and unembedding (5 points)

Describe the role of embedding and unembedding in a transformer model. Make sure to connect this to the values of the model dimension d_{model} , vocabulary size $|\mathcal{V}|$, and sequence length L . State the sizes of relevant matrices and vectors.

Solution.

The embedding transforms a sequence of one-hot token vectors into a representation in which each token is a vector of length d_{model} :

$$E_{L \times d_{\text{model}}} = T_{L \times |\mathcal{V}|} W_E_{|\mathcal{V}| \times d_{\text{model}}},$$

The unembedding transforms the output representation x_{last} of the final sequence position into logits:

$$\text{logits}_{\text{last}} = x_{\text{last}}_{1 \times d_{\text{model}}} W_U_{d_{\text{model}} \times |\mathcal{V}|}.$$

Question 9. Training targets (5 points)

A certain transformer uses lower case letters from the English alphabet as its only tokens. The following sequence is its only training data:

yymfndifhdjsjsjshfgkgjdjdvbkfrbsbivb

The transformer employs a block size of 5, and it is being trained using batches of size 2. One particular batch of training data employs the inputs

$$x = \begin{pmatrix} \text{hdjsj} \\ \text{jd vbk} \end{pmatrix}.$$

For this batch, what value of the targets y should be employed?

Solution.

We need sequences of the training data that match the inputs except shifted right by one position. This gives

$$y = \begin{pmatrix} \text{djsjs} \\ \text{dvbkf} \end{pmatrix}.$$

Question 10. Predictions and loss calculation

A transformer uses the vocabulary $\{\mathbf{a}, \mathbf{b}\}$ with IDs $\{\mathbf{a} : 0, \mathbf{b} : 1\}$. It processes a sequence of length 3, producing the final logits

$$\begin{pmatrix} 3.1 & -0.5 \\ 0.2 & 0.7 \\ -1.1 & -0.1 \end{pmatrix}.$$

(a) **(5 points)** What are the predicted probabilities for the next token, and which token will be output?

Solution.

Prediction uses logits for the final sequence position only, so we compute a softmax of $(-1.1, -0.1)$, yielding

$$\left(\frac{e^{-1.1}}{e^{-1.1} + e^{-0.1}}, \frac{e^{-0.1}}{e^{-1.1} + e^{-0.1}} \right) = (0.27, 0.73).$$

The highest probability corresponds to token ID 1, so the output is **b**.

(b) **(5 points)** Suppose instead that the same logits above were produced by training on a single input sequence $x = \mathbf{aba}$ and target $y = \mathbf{abb}$. What is the contribution of the final token to the cross-entropy loss?

Solution.

The total cross-entropy loss is $-\sum_i \log p_{y_i}$, where y_i is the token ID of the i th target token. The question asks for the contribution from the final token only. The final target is **b** (token ID 1). From the previous question, we know the computed probability of **b** was 0.73. So the contribution of this token to the loss is

$$-\log 0.73 = 0.14.$$

Question 11. Optional. Will not be on the quiz or exam. Single-head attention operator. (0 points)

List the major quantities employed by a single-head attention operator, in the approximate order of computation.

Solution.

token embedding, position embedding, input representation, query vectors, key vectors, value vectors, attention weights, attention-weighted value vectors, projected attention output, post-residual output representation.

See appendix B for some optional additional explanation of this computation.

Total number of points: **60**

Appendix A Optional extra information about transformer computation

Information that appears only in this appendix is not required for quiz or exam questions.

- token embedding = one-hot token matrix $\cdot$ embedding matrix:

$$\underset{L \times d_{\text{model}}}{E_T} = \underset{L \times |\mathcal{V}|}{T} \underset{|\mathcal{V}| \times d_{\text{model}}}{W_E},$$

- position embedding = $\underset{L \times d_{\text{model}}}{E_P}$.
- input representation = token embedding + position embedding:

$$\underset{L \times d_{\text{model}}}{X} = \underset{L \times d_{\text{model}}}{E_T} + \underset{L \times d_{\text{model}}}{E_P}$$

- attention layers iteratively produce output representation X from input representation X :

$$\begin{aligned} X &\rightarrow \text{AttentionLayer}(X) \\ X &\rightarrow \text{AttentionLayer}(X) \\ &\vdots \\ X &\rightarrow \text{AttentionLayer}(X) \end{aligned}$$

- logits = output representation $\cdot$ unembedding matrix:

$$\underset{L \times |\mathcal{V}|}{\text{logits}} = \underset{L \times d_{\text{model}}}{X} \underset{d_{\text{model}} \times |\mathcal{V}|}{W_U}$$

- logits for last token = last row of logits:

$$\underset{1 \times |\mathcal{V}|}{\text{logits}_{\text{last}}} = \underset{L \times |\mathcal{V}|}{\text{logits}}(L - 1, :)$$

- probabilities for last token = softmax of logits for last token:

$$\underset{1 \times |\mathcal{V}|}{p} = \text{softmax} \left(\underset{1 \times |\mathcal{V}|}{\text{logits}_{\text{last}}} \right)$$

- predicted token ID i^* =
 - *either* the token ID with highest probability
 - *or* a token ID chosen randomly according to the calculated probabilities p ,

- *or* a token ID chosen randomly according to probabilities that have been adjusted via a temperature.

That is,

$$\begin{aligned}i^* &= \arg \max_i p_i, \text{ or} \\i^* &= \text{choice}(p), \text{ or} \\i^* &= \text{choice}(p').\end{aligned}$$

The adjusted probabilities p' , for a temperature $\tau > 0$, are given by

$$p'_i = \frac{p_i^{1/\tau}}{\sum_{j=0}^{L-1} p_j^{1/\tau}}.$$

Appendix B Optional extra information about single-head attention operator

Information that appears only in this appendix is not required for quiz or exam questions.

under construction